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ELECTRONIC DEVICE AND METHOD FOR PROCESSING SCALABLE CODEC IN ELECTRONIC DEVICE

Abstract

An electronic device is provided. The electronic device includes a communication circuit, memory, comprising one or more storage media, storing instructions, and one or more processors communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the one or more processors individually or collectively, cause the electronic device to determine information on a scalability applicable to an original image, generate a base layer (BL) image for the original image on the basis of at least one of the resolution, color gamut, gamma, bit depth, or bitrate of the original image, generate at least one artificial intelligence (AI) network on the basis of the original information, the information on the scalability about the original image, and the BL image, compress the BL image and control the communication circuit to transmit a bitstream including the compressed BL image and configuration information on the at least one AI network to an external electronic device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365(c), of an International application No PCT/KR2024/001713, filed on Feb. 6, 2024, which is based on and claims the benefit of a Korean patent application number 10-2023-0017175, filed on Feb. 9, 2023, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2023-0034129, filed on Mar. 15, 2023, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to a technique for processing a scalable codec in an electronic device.

2. Description of Related Art

[0003] A scalable codec may be implemented as, for example, a scalable video codec (SVC) based on H.264 or a scalable high efficiency video coding (SHVC) based on high efficiency video coding (HEVC). Both the SVC and the SHVC may include a base layer (BL) serving as a base and an enhanced layer (EL) for obtaining a final scalable result value based on the BL.

[0004] SVC may be the name of the Annex G extension of the H.264/moving picture experts group-4 (MPEG-4) advanced video coding (AVC) video compression standard. The SVC and/or the SHVC may process encoding of a high-quality video bitstream containing one or more subset bitstreams (e.g., layered coding).

[0005] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

[0006] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide to a technique for processing a scalable codec in an electronic device.

[0007] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0008] In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a communication circuit, memory, including one or more storage media, storing instructions and one or more processors communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the one or more processors individually or collectively, cause the electronic device to determine information on a scalability applicable to an original image, generate a base layer (BL) image for the original image, based on at least one of a resolution, a color gamut, a gamma, a frame rate, a bit depth, or a bit rate of the original image, generate at least one artificial intelligence (AI) network, based on the original image, the information on the scalability for the original image, and the BL image, compress the BL image, and control the communication circuit to transmit a bitstream including the compressed

BL image and configuration information for the at least one AI network to an external electronic device.

[0009] According to an embodiment, the BL image has a different view port from that of an enhanced layer (EL) image. The BL image and the EL image are implemented (or generated) to have a scalability for multiple views.

[0010] In accordance with another aspect of the disclosure, an electronic device is provided. The electronic device includes a communication circuit, memory, including one or more storage media, storing instructions, and one or more processors communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the one or more processors individually or collectively, cause the electronic device to receive, from an external electronic device, a bitstream including a compressed base layer (BL) image and configuration information for at least one artificial intelligence (AI) network, determine at least one scalability based on the configuration information for the at least one AI network, configure at least one AI network for the at least one scalability, and obtain an enhanced layer (EL) image using the BL image and the at least one AI network.

[0011] In accordance with another aspect of the disclosure, a method performed by an electronic device is provided. The method includes determining, by the electronic device, information on a scalability applicable to an original image, generating, by the electronic device, a base layer (BL) image for the original image, based on at least one of a resolution, a color gamut, a gamma, a frame rate, a bit depth, or a bit rate of the original image, generating, by the electronic device, at least one artificial intelligence (AI) network, based on the original image, the information on the scalability for the original image, and the BL image, compressing, by the electronic device, the BL image, and transmitting, by the electronic device, a bitstream including the compressed BL image and configuration information for the at least one AI network to an external electronic device.

[0012] According to an embodiment, the BL image has a different view port from that of an EL image. The BL image and the EL image are implemented (or generated) to have a scalability for multiple views.

[0013] In accordance with another aspect of the disclosure, a method performed by an electronic device is provided. The method includes receiving, by the electronic device, a bitstream including a compressed base layer (BL) image and configuration information for at least one AI network, determining, by the electronic device, at least one scalability based on the configuration information for the at least one artificial intelligence (AI) network, configuring, by the electronic device, at least one AI network for the at least one scalability, and obtaining, by the electronic device, an enhanced layer (EL) image using the BL image and the at least one AI network.

[0014] In accordance with another aspect of the disclosure, one or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by at least one processor of an electronic device individually or collectively, cause the electronic device to perform operations are provided. The operations include determining, by the electronic device, information on a scalability applicable to an original image, generating, by the electronic device, a base layer (BL) image for the original image, based on at least one of a resolution, a color gamut, a gamma, a frame rate, a bit depth, or a bit rate of the original image, generating, by the electronic device, at least one artificial intelligence (AI) network, based on the original image, the information on the scalability for the original image, and the BL image, compressing, by the electronic device, the BL image, and transmitting, by the electronic device, a bitstream including the compressed BL image and configuration information for the at least one AI network.

[0015] According to an embodiment, the BL image has a different view port from that of an EL image. The BL image and the EL image are implemented (or generated) to have a scalability for multiple views.

[0016] In accordance with another aspect of the disclosure, one or more non-transitory computer-

readable storage media storing one or more computer programs including computer-executable instructions that, when executed by at least one processor of an electronic device individually or collectively, cause an electronic device to perform operations are provided. The operations include receiving, by an electronic device, a bitstream including a compressed BL image and configuration information for at least one AI network, determining, by the electronic device, at least one scalability based on the configuration information for the at least one AI network, configuring, by the electronic device, at least one AI network for the at least one scalability, and obtaining an enhanced layer (EL) image using the BL image and the at least one AI network.

[0017] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0019] FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure;

[0020] FIG. 2 illustrates an example of a codec having spatial scalability among scalable video codecs (SVCs) according to an embodiment of the disclosure;

[0021] FIG. 3 illustrates an example of a codec having color gamut scalability among SVCs according to an embodiment of the disclosure;

[0022] FIG. 4 is a diagram illustrating a plurality of AI network-based scalabilities according to an embodiment of the disclosure;

[0023] FIG. 5 illustrates an example of obtaining a final image through an AI network corresponding to each of a plurality of scalabilities according to an embodiment of the disclosure;

[0024] FIG. 6 illustrates an example of obtaining a final image by combining AI networks corresponding to a plurality of scalabilities according to an embodiment of the disclosure;

[0025] FIG. 7 is a diagram illustrating operations of a transmitter and a receiver according to an embodiment of the disclosure;

[0026] FIG. 8 illustrates an example of AI network configuration information in a bitstream according to an embodiment of the disclosure;

[0027] FIG. 9 illustrates an example of AI network configuration information according to an embodiment of the disclosure;

[0028] FIG. 10 illustrates an example of bitstream transmission and interpretation in a transmitter and a receiver according to an embodiment of the disclosure;

[0029] FIG. 11 illustrates an example of a method of pre-transmitting and configuring AI IDs and information about networks in a package according to an embodiment of the disclosure;

[0030] FIG. 12 illustrates an example of AI network configuration and processing in a receiver according to an embodiment of the disclosure;

[0031] FIG. 13 is a diagram illustrating an operation of a transmitter according to an embodiment of the disclosure; and

[0032] FIG. 14 is a diagram illustrating an operation of a receiver according to an embodiment of the disclosure.

[0033] Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.

DETAILED DESCRIPTION

[0034] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

[0035] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

[0036] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0037] It should be appreciated that the blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.

[0038] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP, e.g. a central processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphics processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a wireless fidelity (Wi-Fi) chip, a Bluetooth® chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a fingerprint sensor controller, a display driver integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.

[0039] FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure.

[0040] Referring to FIG. 1, an electronic device **101** in a network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0041] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation.

According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0042] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0043] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0044] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0045] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0046] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0047] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram

device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the strength of force incurred by the touch.

[0048] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0049] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0050] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0051] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0052] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0053] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0054] The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0055] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0056] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module).

A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a fifth generation (5G) network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0057] The wireless communication module **192** may support a 5G network, after a fourth generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the millimeter wave (mmWave) band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0058] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0059] According to an embodiment, the antenna module **197** may form an mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0060] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or

mobile industry processor interface (MIPI)).

[0061] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0062] The electronic device according to embodiments of the disclosure may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0063] It should be appreciated that embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. As used herein, each of such phrases as “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B, or C”, “at least one of A, B, and C”, and “at least one of A, B, or C”, may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with”, “coupled to”, “connected with”, or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0064] As used in connection with embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, logic, logic block, part, or circuitry. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0065] Embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136**

or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0066] According to an embodiment, a method according to embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0067] According to an embodiment, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to an embodiment, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to an embodiment, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0068] In an embodiment of the disclosure, a method of improving bit efficiency by applying an artificial intelligence (AI)-based image processing technology to a scalable video codec (SVC) is proposed. Owing to the bit efficiency improvement, a transmitter may transmit a plurality of scalabilities based on information about a plurality of AI networks to a receiver, and the receiver may selectively use various scalabilities. The transmitter may define configuration information for an AI network and transmit the configuration information for the AI network to the receiver. The receiver may process an image by interpreting the received configuration information for the AI network. In an embodiment of the disclosure, the transmitter may refer to an electronic device that transmits information, a message, and/or a bitstream, and the receiver may refer to an electronic device that receives information, a message, and/or a bitstream.

[0069] In the SVC, the transmitter may keep (or store) an original image to which scalability may be applied, and thus the original image may be used as ground truth in configuring an AI network. The transmitter may transmit configuration information for various AI networks to the receiver. The receiver may configure the same AI network as that of the transmitter based on the AI network configuration information transmitted from the transmitter. The receiver may obtain (or receive) an enhanced layer (EL) image with scalability by inputting a base layer (BL) image to the AI network.

[0070] A scalable codec may be implemented as, for example, a scalable video codec (SVC) based on H.264 or a scalable HEVC (SHVC) based on high efficiency video coding (HEVC). Both the SVC and the SHVC may include a BL and an EL for obtaining a final scalable result value based

on the BL.

[0071] According to an embodiment, scalability may be implemented as spatial scalability, temporal scalability, and/or signal-to-noise ratio (SNR) scalability. According to an embodiment, scalability may be implemented as a hybrid codec for delivering different heterogeneous codecs, and/or color gamut and bit depth scalability for scaling a color gamut and a bit depth.

[0072] Using the SVC technology, a standard dynamic range (SDR) image may be transmitted in the BL and a high dynamic range (HDR) image may be transmitted in the EL. Depending on its decoding capability, the receiver may provide only an SDR image, or when it is capable of HDR processing, the receiver may provide even an HDR image using the EL. For example, in the SVC, the transmitter generates an SDR image corresponding to the BL using an HDR image corresponding to the EL as an original image, and the receiver may generate the HDR image corresponding to the EL again using the SDR image corresponding to the BL provided from the transmitter.

[0073] FIG. 2 illustrates an example of a codec having spatial scalability among SVCs according to an embodiment of the disclosure.

[0074] An electronic device (e.g., the electronic device **101** of FIG. 1) may obtain a scalable bitstream from an input image (or original image) through an EL **210** and a BL **220**.

[0075] The EL **210** may include a decoded picture buffer (DPB), a loop filter, an intra-prediction module, an inverse transform/inverse quantization (IT/IQ) module, a motion estimation/motion compensation (ME/MC) inter-layer prediction module, a transform/quantization (T/Q) module, an entropy encoding module, and an up-sampling module. The BL **220** may include a DPB, a loop filter, an intra-prediction module, an IT/IQ module, an ME/MC inter-layer prediction module, a T/Q module, and an entropy encoding module.

[0076] The electronic device may process the input image (or original image) through the EL **210**, process a down-sampled input image (or original image) through the BL **220**, and input each processing result to a multiplexer (MUX), thereby obtaining a scalable bitstream.

[0077] When the input image (or original image) of the electronic device is a high-resolution image (e.g., ultra high definition (UHD)-level image), the electronic device may apply down-scaling to the input image (or original image) to generate a BL image. The BL image generated by applying down-scaling to the input image (or original image) has a lower resolution (e.g., full high definition (FHD)-level image) than the input image (or original image), and may be encoded independently. The encoded BL image may be implemented as a bitstream into which, for example, the FHD-level image is compressed.

[0078] When encoding the EL image, the electronic device may use an image obtained by up-scaling a restored image (e.g., FHD-level image) of the BL image, as one of reference pictures. The electronic device may use an image obtained by up-scaling the same original image processed in the BL **220** as one of reference pictures in the EL **210**, thereby improving compression efficiency. For example, when the electronic device uses the SVC, the bit efficiency may be improved by approximately 30%, compared to data in which the BL **220** and the EL **210** are separately compressed.

[0079] FIG. 3 illustrates an example of a codec having color gamut scalability among SVCs according to an embodiment of the disclosure.

[0080] BT.2020 (e.g., a wider color gamut than BT.709) corresponding to a UHD TV input **301** may be input to an EL encoding loop **307**, and for example, BT.709 **303** corresponding to SDR may be input to a BL encoding loop **313**.

[0081] An electronic device (e.g., the electronic device **101** of FIG. 1) may obtain a BL reference **311** through the BL encoding loop **313** and perform color space prediction **309** using the BL reference **311**. The electronic device may use a result of the color space prediction **309** and an EL reference **305** in the EL encoding loop **307**. The electronic device may perform bitstream multiplexing **315** based on a result of the BL encoding loop **313** and a result of the EL encoding

loop **307**.

[0082] As in the examples of spatial scalability in FIG. 2 and color gamut scalability in FIG. 3, the SVC may commonly have a reference picture corresponding to a BL (e.g., the BL **220** and the BL encoding loop **313**), and in order to obtain an EL image (e.g., an image processed through an EL) corresponding to scalability, a BL image (e.g., an image processed through the BL) that may be referenced on the time axis may be first decoded and used as one of reference pictures for ME/MC.

[0083] In the SVC, the receiver may selectively decode only the BL image or decode up to the EL image with scalability according to its device capability, thereby selectively performing decoding according to the device capability. For example, when the maximum decoding resolution capability of device A is FHD, the maximum decoding resolution capability of device B is UHD, and a bitstream with spatial scalability using the SVC is encoded into FHD for a BL image and into UHD for an EL image, device A may decode the FHD of the BL image from the same bitstream, and device B may decode up to the EL image to receive the UHD-level image.

[0084] However, when the SVC is used, BL image decoding should always precede to decode the EL image corresponding to scalability, and the BL image should be generated like the EL image corresponding to scalability by actually decoding the BL image in the middle of an EL image decoding process, and used as a reference picture during the encoding of the EL image, which may be complex. When a picture from the BL image is to be used as a reference picture, additional memory may be required.

[0085] In an embodiment of the disclosure, a transmitter (or a first electronic device) may simultaneously transmit BL compressed data (e.g., data compressed in a BL) and information about at least one AI network (or configuration information for the AI network) for scalability to a receiver (or a second electronic device). The receiver may obtain an image corresponding to a desired scalability (e.g., spatial, temporal, SNR, heterogeneous codec, or color gamut) by using the information about the at least one AI network. The transmitter (or the first electronic device) may define information about various AI networks (or configuration information for the AI networks) in a bitstream. The transmitter may transmit a plurality of scalabilities to the receiver by using the information for the AI networks (or configuration information for the AI networks) having a smaller data size than an existing EL image. The receiver may select and/or configure an AI network corresponding to a desired scalability and generate an image of the corresponding scalability.

[0086] FIG. 4 is a diagram illustrating a plurality of AI network-based scalabilities according to an embodiment of the disclosure.

[0087] Referring to FIG. 4, a transmitter **410** may be implemented as an electronic device (e.g., the electronic device **101** of FIG. 1) that transmits information, a message, and/or a bitstream, and a receiver **420** may be implemented as an electronic device (e.g., the electronic device **102** or the electronic device **104** of FIG. 1) that receives information, a message, and/or a bitstream.

[0088] The transmitter **410** and/or the receiver **420** may support one or more scalabilities simultaneously by a scalable codec using a plurality of AI networks **431** to **435**. The receiver **420** may decode a BL image **440** serving as a basis using at least one of the AI networks (at least one of the AI networks **431** to **435**) configured by the transmitter **410**, and obtain a final image (target video) **450** using a configuration of an AI network of a scalability desired by the receiver **420**.

[0089] The transmitter **410** may support transmission and/or reception of images corresponding to a plurality of scalabilities by configuring the plurality of AI networks **431** to **435**. For example, a first AI network **431** may provide and/or support spatial scalability, a second AI network **432** may provide and/or support temporal scalability, a third AI network **433** may provide and/or support SNR scalability, a fourth AI network **434** may provide and/or support HDR scalability, and a fifth AI network **435** may provide and/or support a specific scalability (xxx spatial scalability). The number and/or types of AI networks according to an embodiment of the disclosure are not limited.

[0090] The transmitter **410** may transmit information or configuration information for at least one

of the plurality of AI networks **431** to **435** to the receiver **420**. The transmitter **410** may keep (e.g., store) an original image to which a scalability (spatial scalability) is applied. The transmitter **410** may convert and/or compress the original image into the BL image **440** and transmit it to the receiver **420**.

[0091] According to an embodiment, the receiver **420** may obtain a final image **450** to which spatial scalability **451** is applied based on information about the first AI network **431**. According to an embodiment, the receiver **420** may obtain a final image **450** to which temporal scalability **452** is applied based on information about the second AI network **432**. According to an embodiment, the receiver **420** may obtain a final image **450** to which SNR scalability **453** is applied based on information about the third AI network **433**. According to an embodiment, the receiver **420** may obtain a final image **450** to which HDR scalability **454** is applied based on information about the fourth AI network **434**. According to an embodiment, the receiver **420** may determine at least one scalability to be applied to the final image **450** in response to a user selection or input. According to an embodiment, the receiver **420** may select a plurality of scalabilities in response to a user selection or input and obtain a final image **450** to which the plurality of scalabilities are applied.

[0092] According to an embodiment, the transmitter **410** may include a communication circuit (e.g., the communication module **190** of FIG. **1**) and at least one processor (e.g., the processor **120** of FIG. **1**) that performs at least one operation described above. According to an embodiment, the receiver **420** may include a communication circuit (e.g., the communication module **190** of FIG. **1**) and at least one processor (e.g., the processor **120** of FIG. **1**) that performs at least one operation described above.

[0093] FIG. **5** illustrates an example of obtaining a final image through an AI network corresponding to each of a plurality of scalabilities according to an embodiment of the disclosure.

[0094] Referring to FIG. **5**, a transmitter may be implemented as an electronic device (e.g., the electronic device **101** of FIG. **1**) that transmits information, a message, and/or a bitstream, and a receiver may be implemented as an electronic device (e.g., the electronic device **102** or the electronic device **104** of FIG. **1**) that receives information, a message, and/or a bitstream. The transmitter may include a BL module **510** for processing a BL image and an EL module **520** for processing an EL image. The receiver may include a BL module **530** for processing a BL image and an EL module **540** for processing an EL image.

[0095] According to an embodiment, the transmitter may include a communication circuit (e.g., the communication module **190** of FIG. **1**) and at least one processor (e.g., the processor **120** of FIG. **1**) that performs at least one operation described above. According to an embodiment, the receiver may include a communication circuit (e.g., the communication module **190** of FIG. **1**) and at least one processor (e.g., the processor **120** of FIG. **1**) that performs at least one operation described above.

[0096] The transmitter may apply down-scaling and frame sampling to an original image **501** having the characteristics of UHD 60 fps using the BL module **510** and generate a BL image **503** having the characteristics of FHD 30 fps. The BL image **503** may be compressed to a bit rate (e.g., 5 Mbps) lower than SNR scalability to be decoded later. The receiver may receive a compressed BL image **505** having the characteristics of FHD 30 fps 5 Mbps from the transmitter.

[0097] According to an embodiment, the receiver may apply spatial scalability to the compressed BL image **505** using the EL module **540** and generate a first EL image **507** having the characteristics of UHD 30 fps. According to an embodiment, the receiver may apply temporal scalability to the compressed BL image **505** using the EL module **540** and generate a second EL image **509** having the characteristics of FHD 60 fps. According to an embodiment, the receiver may apply SNR scalability to the compressed BL image **505** using the EL module **540** and generate a third EL image **511** having the characteristics of FHD 30 fps 10 Mbps. According to an embodiment, the receiver may apply spatial and/or temporal scalability to the compressed BL image **505** using the EL module **540** and generate a fourth EL image **513** having the characteristics

of UHD 60 fps. According to an embodiment, the receiver may apply spatial, temporal, and/or SNR scalability to the compressed BL image **505** using the EL module **540** and generate a fifth EL image **515** having the characteristics of UHD 60 fps 80 Mbps. The case of FIG. 5 is an example having an AI network for an EL image corresponding to each final scalability. In this case, information on a scalability from the transmitter may represent information about a combination of a plurality of scalabilities. For example, for UHD@60, spatial and/or temporal may be simultaneously indicated by scalability index information. Information about image characteristics (e.g., resolution, color gamut, gamma, frame rate (fps), bit depth, and/or bit rate) corresponding to an EL image that may be processed in the EL module **540** may be added, and the receiver may select and apply an AI network corresponding to a desired EL image based on the information.

[0098] In an embodiment of the disclosure, without compressing and transmitting an EL image for which final scalability may be created using a BL image, the BL image may be decoded and then each EL image (or result image) may be obtained directly by applying an AI network corresponding to each scalability to the BL image. According to an embodiment, only a resolution and fps characteristics are required for the scalability result image, and since it is not an image in a compressed format, bps characteristics may not be required. However, in the case of SNR scalability, the degree of image quality may be expressed as FHD 30 fps 10 Mbps-level image quality or UHD 60 fps 80 Mbps-level image quality in order to represent the degree of image quality. According to an embodiment, for [in] temporal scalability or [SNR] spatial scalability, the transmitter may set FHD 60 fps down-scaled from the original image, UHD 60 fps in an original scalability as the ground truth of the BL image, FHD 30 fps, when configuring an AI network, and set UHD 60 fps as the ground truth, when configuring an AI network of temporal and spatial scalabilities.

[0099] FIG. 6 illustrates an example of obtaining a final image by a combination of AI networks corresponding to a plurality of scalabilities according to an embodiment of the disclosure.

[0100] Referring to FIG. 6, a transmitter may be implemented as an electronic device (e.g., the electronic device **101** of FIG. 1) that transmits information, a message, and/or a bitstream, and a receiver may be implemented as an electronic device (e.g., the electronic device **102** or the electronic device **104** of FIG. 1) that receives information, a message, and/or a bitstream. The transmitter may include a BL module **610** for processing a BL image and an EL module **620** for processing an EL image. The receiver may include a BL module **630** for processing a BL image and an EL module **640** for processing an EL image.

[0101] According to an embodiment, the transmitter may include a communication circuit (e.g., the communication module **190** of FIG. 1) and at least one processor (e.g., the processor **120** of FIG. 1) that performs at least one operation described above. According to an embodiment, the receiver may include a communication circuit (e.g., the communication module **190** of FIG. 1) and at least one processor (e.g., the processor **120** of FIG. 1) that performs at least one operation described above.

[0102] The transmitter may generate a BL image **603** having the characteristics of FHD 30 fps by applying down-scaling and frame sampling to an original image **601** having the characteristics of UHD 60 fps using the BL module **610**. The BL image **603** may be compressed to a bit rate (e.g., 5 Mbps) lower than SNR scalability to be decoded later. The receiver may receive a compressed BL image **605** having the characteristics of FHD 30 fps 5 Mbps from the transmitter.

[0103] According to an embodiment, the receiver may generate a first EL image **607** having the characteristics of UHD 30 fps by applying spatial scalability to the compressed BL image **605** using the EL module **640**. According to an embodiment, the receiver may generate a second EL image **609** having the characteristics of FHD 60 fps by applying temporal scalability to the compressed BL image **605** using the EL module **640**, and generate a third EL image **611** having the characteristics of UHD 60 fps by applying spatial scalability to the second EL image **609**.

[0104] According to an embodiment, the receiver may generate a fourth EL image **613** having the

characteristics of FHD 30 fps 10 Mbps by applying SNR scalability to the compressed BL image **605** using the EL module **640**. According to an embodiment, the receiver may generate a fifth EL image **615** having the characteristics of UHD 30 fps by applying spatial scalability to the fourth EL image **613** using the EL module **640**. According to an embodiment, the receiver may generate a sixth EL image **617** having the characteristics of UHD 60 fps 80 Mbps by applying temporal scalability to the fifth EL image **615** using the EL module **540**.

[0105] Since the image corresponding to the BL in the receiver has the characteristics of FHD 30 fps 5 Mbps, for example, the AI network corresponding to temporal scalability may be used to obtain an image of UHD 60 fps, and then the AI network corresponding to spatial scalability may be used again for the resulting FHD 60 fps to obtain a final image of UHD 60 fps. Or, conversely, spatial scalability may be applied first to create UHD 30 fps, and then temporal scalability may be applied to create UHD 60 fps. The order that may lead to better results may be set or the receiver may choose between the two orders. In an embodiment, the AI network used for spatial scalability of UHD 30 fps may be reused as the AI network for spatial scalability, if it is created by considering even a scalability combination in the transmitter, and a new AI network for spatial scalability may be additionally transmitted from the transmitter to the receiver. When the receiver needs to implement an FHD 60 fps image instead of a UHD 60 fps image or even when the receiver does not have a device capability of processing or displaying up to a UHD image, the receiver may implement the FHD 60 fps image by applying temporal scalability.

[0106] FIG. 7 is a diagram illustrating operations of a transmitter and a receiver according to an embodiment of the disclosure.

[0107] Referring to FIG. 7, the transmitter may be implemented as an electronic device (e.g., the electronic device **101** of FIG. 1) that transmits information, a message, and/or a bitstream, and the receiver may be implemented as an electronic device (e.g., the electronic device **102** or the electronic device **104** of FIG. 1) that receives information, a message, and/or a bitstream.

[0108] According to an embodiment, the transmitter may include a communication circuit (e.g., the communication module **190** of FIG. 1) and at least one processor (e.g., the processor **120** of FIG. 1) that performs at least one operation described above. According to an embodiment, the receiver may include a communication circuit (e.g., the communication module **190** of FIG. 1) and at least one processor (e.g., the processor **120** of FIG. 1) that performs at least one operation described above.

[0109] Referring to FIG. 7, in operation **701**, the transmitter may identify and/or determine an original image (or original video). In operation **703**, the transmitter may determine information about at least one scalability applicable to the original image. In operation **705**, the transmitter may determine a BL image for the original image based on at least one of a resolution, a color gamut, a gamma, a frame rate (fps), a bit depth, or a bit rate. In operation **707**, the transmitter may generate at least one AI network corresponding to the determined information about the at least one scalability. According to an embodiment, the transmitter may generate at least one AI network based on at least one of the original image, the information about the at least one scalability for the original image, or the BL image. In operation **709**, the transmitter may compress the BL image and generate a bitstream including the compressed BL image and configuration information for the at least one AI network. In operation **711**, the transmitter may transmit the bitstream including the compressed BL image and the configuration information for the at least one AI network to the receiver. According to an embodiment, the BL image may have a different view port from an EL image. The BL image and the EL image may be implemented (or generated) to have scalability for multiple views.

[0110] In operation **713**, the receiver may identify a scalability identifier (ID) in the bitstream received in operation **711**. According to an embodiment, the scalability ID may be included in the configuration information for the AI network. In operation **715**, the receiver may determine at least one scalability to be used to obtain a final image. In operation **717**, the receiver may configure at

least one AI network for the determined at least one scalability. The receiver may configure an AI network of the same type as that of the transmitter by using syntaxes in the configuration information (metadata) for the AI network. The receiver may decode the BL image in operation 719, and pass (or apply) the decoded BL image through the at least one AI network in operation 721. In operation 723, the receiver may obtain an EL image based on a result of operation 721. [0111] FIG. 8 illustrates an example of AI network configuration information in a bitstream according to an embodiment of the disclosure.

[0112] Referring to FIG. 8, the AI network configuration information (or metadata) may include syntax information such as information about a scalability type, information about an AI network configuration, and/or information about operators in the AI network configuration.

[0113] The information about the scalability type may include at least one scalability ID. The scalability ID may be any one of spatial, temporal, SNR, and HDR.

[0114] The information about the AI network configuration may include at least one of the number of channels, a layer configuration, or a filter for each layer. The number of channels may include information about the number of input channels and/or the number of output channels, the layer configuration may include information about the number of layers and/or filter information, and the filter for each layer may include information about a filter type, a convolution size, and/or a convolution coefficient.

[0115] The information about the operators in the AI network configuration may include at least one of filter information, an activation function, or an optimizer. The filter information may include information about at least one of convolution, Gaussian, median, or morphology. The activation function may include at least one of Relu, Sigmoid, or Hyperbolic Tangent (tanh). The optimizer may include at least one of stochastic gradient decent (SGD), Momentum, RMSprop, or Adam.

[0116] The AI network configuration information (or metadata) may be information about an AI network to be used or configured in the transmitter and receiver. The AI network configuration information (or metadata) may include information about the type of an AI network used for each frame, when receiving actual image data information, and a parameter value (e.g., filter coefficient) used in the AI network.

[0117] FIG. 9 illustrates an example of AI network configuration information according to an embodiment of the disclosure.

[0118] Referring to FIG. 9, the transmitter may transmit AI network configuration information corresponding to two scalabilities to the receiver. FIG. 9 illustrates an example of AI network configuration information in the form of a table, for 1) a case where the scalability ID is HDR (HDR scalability) and 2) a case where the scalability ID is Temporal (temporal scalability). The AI network configuration information illustrated in FIG. 9 is an example for the convenience of description, to which the technical idea of the disclosure is not limited.

[0119] For example, in the AI network configuration information for HDR scalability in FIG. 9, the scalability ID may be set to HDR, the number of layers may be set to 2, the number of input channels may be set to 3, and the number of output channels may be set to 2. Layer1 configuration may be set to linear, Layer1 linear filter may be set to conv, layer1 Conv kernel size W may be set to 3, layer1 Conv kernel size H may be set to 3, layer1 Conv number may be set to 5, and for each of Layer 1 Conv1 coefficient to Layer 1 Conv 5 coefficient, 9 coefficients may be defined. Layer2 configuration may be set to non-linear, and Layer2 linear filter may be set to Relu.

[0120] For example, in the AI network configuration information for temporal scalability in FIG. 9, the scalability ID may be set to temporal, the number of layers may be set to 6, the number of input channels may be set to 3, the number of output channels may be set to 9, and Layer 1 configuration may be set to linear.

[0121] According to an embodiment, when a plurality of layers are configured in AI network configuration information, whether it is a skip connection or a residual connection may be added in a syntax, and in the case of the skip connection, layer indexes where skipping starts and ends may

be added.

[0122] FIG. **10** illustrates an example of bitstream transmission and interpretation in a transmitter and a receiver according to an embodiment of the disclosure.

[0123] Referring to FIG. **10**, a bitstream of a plurality of scalable codecs may include sequence parameter set (SPS) data, picture parameter set (PPS) data, a plurality of supplemental enhancement information (SEI) data, and a plurality of slice data.

[0124] According to an embodiment, the plurality of SEI data may include a first SEI data SEI-0, a second SEI data SEI-1, and a third SEI data SEI-2. The first SEI data SEI-0 may include configuration information for a first AI network (AI Network information 0), the second SEI data SEI-1 may include configuration information for a second AI network (AI Network information 1), and the third SEI data SEI-2 may include configuration information for a third AI network (AI Network information 2).

[0125] The transmitter may transmit the bitstream of the plurality of scalable codecs to the receiver. According to an embodiment, the receiver may configure the first AI network based on the configuration information for the first AI network (AI Network information 0) included in the bitstream. According to an embodiment, the receiver may configure the second AI network based on the configuration information for the second AI network (AI Network information 1) included in the bitstream. According to an embodiment, the receiver may configure the third AI network based on configuration information for the third AI network (AI Network information 2) included in the bitstream.

[0126] According to an embodiment, the receiver may select at least one of a plurality of scalability IDs using the configuration information for at least one AI network and compressed data of a BL image transmitted from the transmitter, and configure the same AI network as that of the transmitter. According to an embodiment, the receiver may decode the BL image and pass the decoded image through each AI network configured for each scalability, thereby obtaining an image of a scalability to be finally obtained. Alternatively, as described with respect to FIG. **6**, the receiver may obtain an image of a scalability to be finally obtained by a combination of a plurality of scalabilities.

[0127] According to an embodiment, when the receiver has a limited device capability or performance is satisfied only with the BL image, the receiver may obtain an EL image using only the BL image.

[0128] According to an embodiment, the receiver may configure at least one AI network by pre-reading AI network information at a position independent of that of encoded frame data in the bitstream.

[0129] FIG. **11** illustrates a method of pre-transmitting and configuring AI IDs and information about networks in a package according to an embodiment of the disclosure.

[0130] While FIG. **10** illustrates a method of transmitting all configuration information for an AI network transmitted together in a bitstream of an image, FIG. **11** illustrates a method of transmitting a package corresponding to an AI network before a bitstream of an image, in consideration of a delay time involved in configuring the AI network in the receiver.

[0131] Referring to FIG. **11**, the transmitter may transmit an AI network package including configuration information for a plurality of AI networks, AI Network 0 to AI Network 4 to the receiver before transmitting a bitstream of an image. The receiver may pre-configure at least one AI network using the pre-received AI network package.

[0132] According to an embodiment, when the transmitter pre-transmits the AI network package to the receiver, the transmitter may not transmit AI network configuration information separately, when transmitting the bitstream of the image. According to an embodiment, in the case where the transmitter pre-transmits the AI network package to the receiver, when transmitting the bitstream of the image, the transmitter may transmit, to the receiver, an AI network index indicating information on a scalability, for each scalability, and a coefficient value and/or a weight to be used in an AI

network as information about each AI network as in the example of FIG. 8.

[0133] According to an embodiment, the transmitter may identify whether the configuration of the transmitted AI network package is fully prepared in the receiver through communication with the receiver, and transmit the bitstream of the image at a time after the identification. According to an embodiment, a single AI network for each scalability may be fixed throughout the image and updated in the middle of an image sequence.

[0134] According to an embodiment, considering a time required for the receiver to configure the updated AI network, the transmitter may transmit the configuration information for the AI network to the receiver before frame data to be applied to the updated network. The configuration information for the AI network may include information indicating a frame up to which the existing AI network should be applied and/or a frame from which the new AI network should be applied. The information indicating the frame from which the new AI network is to be applied may be a decoding order or a display order. According to an embodiment, the new AI network may be predefined to be applied from an N^{sup}.th frame or later after a current frame.

[0135] FIG. 12 illustrates an example of configuring and processing an AI network in a receiver according to an embodiment of the disclosure.

[0136] FIG. 12 illustrates an example of obtaining a final image (or EL image) corresponding to a scalability after configuring an AI network using a received bitstream in a receiver, as in FIG. 10.

[0137] A bitstream of a plurality of scalable codecs transmitted from the transmitter to the receiver may include SPS data, PPS data, a plurality of SEI data, and a plurality of slice data.

[0138] The receiver may decode a BL image based on the bitstream of the plurality of scalable codecs. For example, when the BL image is an FHD 30 fps image, the receiver may configure a first AI network, AI Network-0 for spatial scalability and/or a second AI network, AI Network-1 for temporal scalability by interpreting the received bitstream. The receiver may obtain an EL image of UHD 30 fps by passing a decoded BL image of FHD 30 fps through the first AI network, AI Network-0. The receiver may obtain an EL image of FHD 60 fps by passing the decoded BL image of FHD 30 fps through the second AI network, AI Network-1. Alternatively, as in the example of FIG. 6, a final EL image of UHD 60 fps may be obtained by passing the image through the second AI network after passing through the first AI network. The transmitter may additionally transmit additional information indicating whether a plurality of AI networks may be used in combination in this manner, and as described before, in order to ensure image quality performance in this case, learning may also be performed in a combined state, when the transmitter generates the corresponding network, so that each AI network may be effective even in the combined form. Alternatively, as in the example of FIG. 5, each independent AI network may be transmitted to obtain the final EL image.

[0139] FIG. 13 is a diagram illustrating an operation of a transmitter according to an embodiment of the disclosure.

[0140] The transmitter (e.g., the electronic device 101 of FIG. 1) may include a communication circuit (e.g., the communication module 190 of FIG. 1) and at least one processor (e.g., the processor 120 of FIG. 1) that performs at least one operation described above.

[0141] Referring to FIG. 13, in operation 1310, the transmitter (e.g., the electronic device 101 of FIG. 1) may determine information on a scalability for an original image, to be transmitted to an external electronic device (e.g., the electronic device 102 or the electronic device 104 of FIG. 1) using the at least one processor (e.g., the processor 120 of FIG. 1). In operation 1320, the transmitter may determine a BL image for the original image, based on at least one of a resolution, a color gamut, a gamma, a frame rate, a bit depth, or a bit rate using the at least one processor (e.g., the processor 120 of FIG. 1). In operation 1330, the transmitter may generate at least one AI network based on the original image, the information on the scalability for the original image, and the BL image, using the at least one processor (e.g., the processor 120 of FIG. 1). In operation 1340, the transmitter may transmit a bitstream including a compressed BL image and configuration

information for the at least one AI network to the external electronic device, using the communication circuit (e.g., the communication module **190** of FIG. **1**). According to an embodiment, the BL image may have a different view port from an EL image. The BL image and the EL image may be implemented (or generated) to have scalability for multiple views.

[0142] According to an embodiment, the configuration information for the at least one AI network may include at least one of information about a scalability ID, information about an AI network configuration, or information about operators in the AI network configuration. According to an embodiment, the information about the AI network configuration may include at least one of the number of channels, a layer configuration, or a filter for each layer. According to an embodiment, the information about the operators in the AI network configuration may include at least one of filter information, an activation function, or an optimizer.

[0143] According to an embodiment, the bitstream may include SPS data, PPS data, and a plurality of SEI data. According to an embodiment, each of the plurality of SEI data may include configuration information for a different AI network. According to an embodiment, the scalability may be any one of spatial scalability, temporal scalability, SNR scalability, a hybrid codec, color gamut scalability, and HDR scalability.

[0144] FIG. **14** is a diagram illustrating an operation of a receiver according to an embodiment of the disclosure.

[0145] The receiver may include a communication circuit (e.g., the communication module **190** of FIG. **1**) and at least one processor (e.g., the processor **120** of FIG. **1**) that performs at least one operation described above.

[0146] Referring to FIG. **14**, in operation **1410**, the receiver (e.g., the electronic device **101** of FIG. **1**) may receive a bitstream including a compressed BL image and configuration information for at least one AI network from an external electronic device (e.g., the electronic device **102** or the electronic device **104** of FIG. **1**), using the communication circuit (e.g., the communication module **190** of FIG. **1**). In operation **1420**, the receiver may determine at least one scalability based on the configuration information for the at least one AI network, using the at least one processor (e.g., the processor **120** of FIG. **1**). In operation **1430**, the receiver may configure at least one AI network for the at least one scalability, using the at least one processor (e.g., the processor **120** of FIG. **1**). In operation **1440**, the receiver may obtain an EL image using the BL image and the at least one AI network through the at least one processor (e.g., the processor **120** of FIG. **1**).

[0147] According to an embodiment, in a computer-readable storage medium storing at least one instruction, the at least one instruction may, when executed by at least one processor, cause an electronic device to perform a plurality of operations. The plurality of operations may include determining information on a scalability applicable to an original image, to be transmitted to an external electronic device. The plurality of operations may include determining a BL image for the original image, based on at least one of a resolution, a color gamut, a gamma, a frame rate, a bit depth, or a bit rate. The plurality of operations may include generating at least one AI network, based on the original image, the information on the scalability for the original image, and the BL image. The plurality of operations may include transmitting a bitstream including a compressed BL image and configuration information for the at least one AI network to the external electronic device.

[0148] According to an embodiment, in a computer-readable storage medium storing at least one instruction, the at least one instruction may, when executed by at least one processor, cause an electronic device to perform a plurality of operations. The plurality of operations may include receiving, from an external electronic device, a bitstream including a compressed BL image and configuration information for at least one AI network. The plurality of operations may include determining at least one scalability based on the configuration information for the at least one AI network. The plurality of operations may include configuring at least one AI network for the at least one scalability. The plurality of operations may include obtaining an EL image using the BL

image and the at least one AI network.

[0149] In the disclosure, there is no limitation on the AI network configuration, and various AI networks may be implemented according to design specifications. According to an embodiment, since network complexity may increase due to the freedom of configuring an AI network, the transmitter and/or the receiver may pre-set a list of a plurality of AI networks. According to an embodiment, the transmitter (or the receiver) may select a suitable AI network from the list of the plurality of AI networks and transmit selection information to the receiver (or the transmitter). According to an embodiment, the selection information may simply include the index of a corresponding AI network. According to an embodiment, the transmitter and/or the receiver may modify information about some networks after selecting the AI network.

[0150] In the disclosure, SVC-based images have been mainly described for convenience of description, but the technical idea of the disclosure may also be applied to still images (e.g., JPEG, BMP, HEIF, etc.) and/or GIF. According to an embodiment, an image corresponding to an additional scalability may be obtained by selectively passing a still image as a basis, such as JPEG, through an AI network, and in addition to the afore-described spatial, temporal, SNR, gamut and bit-depth, HDR, depth-based outfocusing, and various filter effects may be obtained as a result image. According to an embodiment, in the case of a still image, a compressed or uncompressed original still image in a bitstream may be used as a BL image.

[0151] According to an embodiment, AI network configuration information may be transmitted at a specific position before actual image information in a bitstream of an image. According to an embodiment, at least one AI network package including the AI network configuration information may be first transmitted from the transmitter to the receiver, and the receiver may configure an AI network using the previously received AI network package. Thereafter, the receiver may obtain a result image for a final scalability after network processing using network index information and other information to be used (a coefficient value, a weight value, and so on) for a network, transmitted in the bitstream of the image. The values corresponding to the other information may be updated in the bitstream, as needed.

[0152] The embodiments of the disclosure are not limited to scalable codecs, and may also be applied when AI network information is transmitted from a transmitter to a receiver and the receiver restores and/or uses an AI network.

[0153] According to an embodiment, an electronic device (**101** of FIG. **1**) may include a communication circuit (**190** of FIG. **1**), and a processor (**120** of FIG. **1**) connected to the communication circuit (**190** of FIG. **1**). The at least one processor (**120** of FIG. **1**) may generate at least one AI network for a target image. The at least one processor (**120** of FIG. **1**) may transmit configuration information for the at least one AI network to an external electronic device, so that the at least one AI network for the target image is restored in the external electronic device.

[0154] According to an embodiment, an electronic device (**101** of FIG. **1**) may include a communication circuit (**190** of FIG. **1**); and at least one processor (**120** of FIG. **1**) connected to the communication circuit (**190** of FIG. **1**). The at least one processor (**120** of FIG. **1**) may receive configuration information for at least one AI network for a target image from an external electronic device. The at least one processor (**120** of FIG. **1**) may restore the at least one AI network for the target image based on the configuration information for the at least one AI network.

[0155] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

- 1.** An electronic device comprising: a communication circuit; memory, comprising one or more storage media, storing instructions; and one or more processors communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the one or more processors individually or collectively, cause the electronic device to: determine information on a scalability applicable to an original image, generate a base layer (BL) image for the original image, based on at least one of a resolution, a color gamut, a gamma, a frame rate, a bit depth, or a bit rate of the original image, generate at least one artificial intelligence (AI) network, based on the original image, the information on the scalability for the original image, and the BL image, and compress the BL image, and control the communication circuit to transmit a bitstream including the compressed BL image and configuration information for the at least one AI network to an external electronic device.
- 2.** The electronic device of claim 1, wherein the configuration information for the at least one AI network includes at least one of information on a scalability ID, information on an AI network configuration, or information on operators in the AI network configuration.
- 3.** The electronic device of claim 2, wherein the information on the AI network configuration includes at least one of a number of channels, a layer configuration, or a filter for each layer.
- 4.** The electronic device of claim 2, wherein the information on the operators in the AI network configuration includes at least one of filter information, an activation function, or an optimizer.
- 5.** The electronic device of claim 1, wherein the bitstream includes sequence parameter set (SPS) data, picture parameter set (PPS) data, and one or more supplemental enhancement information (SEI) data.
- 6.** The electronic device of claim 1, wherein the scalability is any one of spatial scalability, temporal scalability, signal-to-noise ratio (SNR) scalability, a hybrid codec, color gamut scalability, and high dynamic range (HDR) scalability.
- 7.** An electronic device comprising: a communication circuit; memory, comprising one or more storage media, storing instructions; and one or more processors communicatively coupled to the communication circuit and the memory, wherein the instructions, when executed by the one or more processors individually or collectively, cause the electronic device to: receive, from an external electronic device, a bitstream including a compressed base layer (BL) image and configuration information for at least one artificial intelligence (AI) network, determine at least one scalability based on the configuration information for the at least one AI network, configure at least one AI network for the at least one scalability, and obtain an enhanced layer (EL) image using the BL image and the at least one AI network.
- 8.** The electronic device of claim 7, wherein the configuration information for the at least one AI network includes at least one of information on a scalability ID, information on an AI network configuration, or information on operators in the AI network configuration.
- 9.** The electronic device of claim 8, wherein the information on the AI network configuration includes at least one of a number of channels, a layer configuration, or a filter for each layer.
- 10.** The electronic device of claim 8, wherein the information on the operators in the AI network configuration includes at least one of filter information, an activation function, or an optimizer.
- 11.** The electronic device of claim 7, wherein the bitstream includes sequence parameter set (SPS) data, picture parameter set (PPS) data, and one or more supplemental enhancement information (SEI) data.
- 12.** The electronic device of claim 7, wherein the scalability is any one of spatial scalability, temporal scalability, signal-to-noise ratio (SNR) scalability, a hybrid codec, color gamut scalability, and high dynamic range (HDR) scalability.
- 13.** A method performed by an electronic device, the method comprising: determining, by the electronic device, information on a scalability applicable to an original image; generating, by the electronic device, a base layer (BL) image for the original image, based on at least one of a

resolution, a color gamut, a gamma, a frame rate, a bit depth, or a bit rate of the original image; generating, by the electronic device, at least one artificial intelligence (AI) network, based on the original image, the information on the scalability for the original image, and the BL image; and compressing, by the electronic device, the BL image, and transmitting, by the electronic device, a bitstream including the compressed BL image and configuration information for the at least one AI network.

14. The method of claim 13, wherein the configuration information for the at least one AI network includes at least one of information on a scalability ID, information on an AI network configuration, or information on operators in the AI network configuration.

15. The method of claim 14, wherein the information on the AI network configuration includes at least one of a number of channels, a layer configuration, or a filter for each layer.

16. The method of claim 14, wherein the information on the operators in the AI network configuration includes at least one of filter information, an activation function, or an optimizer.

17. The method of claim 13, wherein the bitstream includes sequence parameter set (SPS) data, picture parameter set (PPS) data, and one or more supplemental enhancement information (SEI) data.

18. The method of claim 13, wherein the scalability is any one of spatial scalability, temporal scalability, signal-to-noise ratio (SNR) scalability, a hybrid codec, color gamut scalability, and high dynamic range (HDR) scalability.
